

How do people in the UK and Japan imagine an encounter with a robot?: a story-stem study

Maki Rooksby*

Social Brain in Action (SOBA) Lab, School of Psychology
and Neuroscience
University of Glasgow
Glasgow, United Kingdom
maki.rooksby@glasgow.ac.uk

Yumiko Muto

Brain Science Institute
Tamagawa university
Machida, Tokyo, Japan
muto@lab.tamagawa.ac.jp

Terry J Goetz

Social Brain in Action (SOBA) Lab, School of Psychology
and Neuroscience
University of Glasgow
Glasgow, United Kingdom
terrygoetz2@yahoo.com

Emily S. Cross

ETH Zurich
Zurich, Switzerland
University of Glasgow
Glasgow, United Kingdom
ecross@ethz.ch

Abstract

The role of culture for human–robot interaction (HRI) remains largely inconclusive. Yet, experimental investigations are challenging, as findings are often constrained by specific experimental setups. A qualitative approach—one where people produce their own HRI situations—may offer a unique insight into cultural factors during HRI. Using a story-stem technique, people who identified themselves as UK or Japanese nationals were asked to write a short story, beginning with a fellow human of their country group encountering a robot. Thematic analysis focusing on the nature of human–robot relationship as well as the extent to which the robot character is portrayed to be agentic, indicated that stories enacting both highly agentic robot characters and an in-group relationship were far more frequent in the Japanese group. The findings are discussed as to what contribution such a qualitative approach may make for the ongoing work to design culturally sensitive robots.

CCS Concepts

• **Social and professional topics** → **Cultural characteristics**; • **Theory of computation** → *Semantics and reasoning*.

Keywords

human-robot interaction, cross-cultural analysis, story-stem method

ACM Reference Format:

Maki Rooksby, Terry J Goetz, Yumiko Muto, and Emily S. Cross. 2025. How do people in the UK and Japan imagine an encounter with a robot?: a story-stem study. In *Extended Abstracts of the CHI Conference on Human Factors in*

*Corresponding author. Maki Rooksby is now at the College of Arts and Humanities, University of Glasgow, and J.T. Goetz is now at University of Staffordshire.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720127>

Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan.
ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3706599.3720127>

1 Introduction

Role of culture has been recognised a key to understanding human experience of interaction with robots, in particular, for so-called social robots which task is to communicate with humans. At the same time, it has been known that culture is a rather slippery and complex subject to study. Edward Hall [4] used the metaphor of a tip of an iceberg to describe this challenge: much about culture is often hidden underneath our overt behaviour, or even from our own consciousness. Published cross-cultural analyses within the Human–robot interaction (HRI) literature have often employed behavioural as well as experimental approaches; these studies typically group people by nationality or country and compare them under a given situation in which to encounter or interact with a given robot. Some of such studies have reported group-level differences, for example, Japanese people have been reported to be positive about robots or more readily accepting of robots than US or European people [8]. At the same time, there is a sense that as the body of studies grow, so do discrepancies across findings, even when the same nations are studied [3][7]. While such inquiries as to what may separate some groups from others are themselves warranted and ongoing, an equally valuable question may be one that is grounded within the above challenge of studying culture itself, and concerns how we approach and study culture in HRI research. Here, an avenue that may offer a unique perspective may be to deliberately remove people from a physical and live encounter with a robot, and to shift the context of study to people's imaginative world. What if, people were to be asked to write a story, starting with a fairly unspecified encounter with a robot? What might their imagination of that encounter be like? What might they imagine for the robot to look like or its capabilities; what sort of relationship, if any, may be imagined developing between human and robot characters? Such imaginations of human–robot encounters would, in narrative format, by its nature, be suited for qualitative analysis, which may uniquely complement the ongoing research dominated by quantitative or experimental methods. While such an

approach may not directly address or help resolve the inconsistencies in the current literature, it may in the least, offer a brief insight into our primitive assumptions or attitudes that may underpin the behaviours that surface in experimental research. Such an approach may also effectively control for the influence of social desirability in laboratory studies; or any other limitations such as a lack of choice for experimental designs due to robot availability or capability. This was the premise from which the current study was designed and carried out.

1.1 Context and aims

The study was part of a larger UK-Japan cross-cultural investigation of HRI and was designed to complement two other ongoing studies at the time, both of which took experimental and quantitative designs. In line with the overall investigation, the focus on the UK-Japan cross-cultural analysis was motivated by the similarities and differences between the two countries: for example, both are island nations, sharing a strong tradition in tea culture or gardening, yet one is in Europe the other in Asia, thus each under distinct cultural traditions. Most crucially for HRI research, while Japan is frequently selected for cross-cultural studies [7], the UK is much less represented in this context. A case of UK-Japan comparison could contribute a critical analysis as to whether narratives of Japanese people indeed suggest 'robot mania' [8] as well as help address an under-representation in the HRI literature.

The specific objectives of this particular study were two-fold; 1) to gain qualitative insight into what might happen in people's minds when they see a robot in their surroundings; 2) to study the ways in which cultural and contextual signals contained in the narrative are used and woven into stories by UK and Japanese participants, and to study whether the two cultural groups present culturally specific ways in which the stories are constructed.

1.2 Story-stem method

The study employed a story-stem method [1] where a short beginning, 'a stem', of a story is presented to the participants, who are then invited to continue and complete the story as led by their imaginations inspired by the stem. Originally developed as a projective technique in psychoanalysis and psychotherapy, the crux of the story stem method is to present people with a short beginning with a degree of ambiguity, to act as a prompt; this is considered to elicit emotions or thoughts that are often masked, including unconscious ideas or assumptions influenced by social desirability or social norms in the minds of participants. It has been shown a valid and effective research tool to study attachment relationships in children or unconscious mental constructs in adults, such as stereotypes or racial prejudices. A small body of existing research within the HRI literature has deployed the method, including studies of what people may imagine for an encounter with a sex robot [13] [12]. These have richly informed human vision of a potential future with sex robots: in the narratives generated in the study, robots were portrayed as a highly efficient (sex) worker/machine, commercially valuable, and capable of empathy and feelings to an uncanny extent. The benefit of the technique was thus considered pertinent and effective for the current study and its objective: to gain insight into how two groups of people, one with predominant

cultural experience in the UK and the other in Japan, may imagine a human character's encounter with a robot, and how the narrative can (or cannot) develop any relationships with the robot, as well as what functions or abilities of the robot may be portrayed in the narratives. The stem involved a human and a robot. The human was named to match the cultural group of participants that will complete the stem (i.e., an English name for the UK group; a Japanese name for the Japanese group). The rationale for the alignment was to prompt participants to imagine an encounter between a robot and a fellow human from their cultural group (an in-group member), so that to activate their sense of cultural identity as they developed their narrative.

1.3 Research questions

Two specific questions addressed in the study concerned: 1) narrative descriptions of human and robot characters: where the human is a fellow member of their cultural group, how do people portray a relationship, if any, between the human and the robot? Would it be a friendly or antagonistic relationship, if any relationship develops in the story?; 2) assignment of agency to the robot character—would the robot be portrayed as an agent rather than an object? We aimed to contribute to the literature by investigating whether Japanese and UK people differed in the way they granted agency to the robotic character. Much debate is ongoing around agency in the HRI literature, including types of agency or its respective definitions, particularly over social agent [6]. Given the aim of the current study, of studying what people imagine for a human–robot encounter via creative writing, our analysis of agency was led by the stories collected rather than by an *a priori* definition: agency was considered in the broadest manner possible; any descriptions that referred to internal processes of an entity (a character in stories), such as thoughts, intentions or emotions as defined by Dennett [2]; in addition, we also included in our analysis those broader descriptions that implied treatment of the robot as a being rather than an object: for example, use of personal pronouns, or other descriptions of personification. A questionnaire presented to participants after the story writing task asked participants to provide information on their previous exposure to robots as well as other demographic information to provide background data to support the analysis. The study was preregistered on the Open Science Framework (OSF) [10].

2 METHODS

The study was built and delivered as an online survey (using Qualtrics): in English for the UK and Japanese for the Japanese participants. It was designed as a stand-alone participation session, which guided participants through a process of informed consent, leading to the narrative writing task, and finishing with a series of short demographic and background questionnaires. The survey was first constructed, and the text finalized in English by M.R. and J.T.G., and M.R. translated the text into Japanese. Reverse translation (Japanese back to the English) was provided by Y.M., another bilingual author; Y.M.'s translation matched the English stem, and the instructions were only differed in minor grammatical aspects. The College of Science and Engineering Ethics Committee at the University of

Glasgow reviewed and issued an ethical approval for the study (application number: 30021001).

2.1 Participants

Participants were first recruited on Prolific. While the platform was effective for the UK group, it was not for Japanese participants. As a result, an advert was posted on a Japanese online recruitment space for research studies, Jikken-baito, which advertised our study on their website as well as via their accounts on social media channels. Eligibility criteria is listed below.

- Participants considered themselves British/Japanese to take part in the UK and Japan respectively.
- Lived in the UK/Japan for most of their adult life, i.e., where they have lived elsewhere, the total period of it was within 6 months.
- Sufficiently proficient in English/Japanese to compose a short story and to complete the survey.

While recruitment relied upon self-declaration by the participants, their responses to eligibility questions were carefully and promptly studied. Informed consent included the agreement to receive inquiries from the research team regarding any ambiguity or inconsistency. All the enquiries were satisfactorily resolved by follow up contact. Following the target sample size specified in the OSF preregistration (30-40 people in each country), we collected 41 and 44 valid sets of responses for the UK and Japanese groups respectively. Participants received a token of appreciation for their time via the platform where they were recruited, approximately £3 via Prolific or 500-yen worth of an online gift card. The survey was designed to be completed in around 30 minutes; the participants were informed of the timescale and of the above payment at the point of recruitment; however, some were faster while others spent longer completing their responses.

2.2 Tasks and instructions

2.2.1 The story-writing task. As the most pertinent component of the study, the story-writing task was presented immediately after participants confirmed their consent to take part. Instruction for the story-writing task was as follows (for the UK participants):

“The following sentence is the beginning of a story about a human and a robot:

A robot is facing Sam from a few metres away...

Your task is to imagine what might happen next and to complete the story in the box below.

Please complete the story with either Sam or the robot as the main character. The choice of main character, what the main character is like, as well as what happens in this story, are completely up to you!”

A gender-neutral name, Sam was selected for the human character to enable freedom of imagination for the UK participants; as an equivalent, Chihiro, was used for the Japanese participants. The gender-neutral names were used so that participants had an opportunity to assign a gender identity for the human character, if wished. The rationale behind the naming was also to instil a sense of an in-group identification with participants, to ensure there was at least one human of their own cultural group in the story. In the

UK, participants were instructed to aim their story to be between 100 and 500 words. As an approximate equivalent, Japanese participants were asked to aim at writing between 200 and 1000 characters using the three writing systems of the Japanese language. A guide was provided in line number for participants in both countries: 8-9 lines for the UK, and up to 10 lines for the Japanese group inside an identical-size text box provided. A step was built in to ask participants to read through their story and to confirm it was completed. At this point, participants were also informed that the research team may wish to contact them at a later date to check their understanding of the story. An agreement in principle was established as an additional and optional consent, which served as an additional quality assurance measure as recommended in the literature [11].

2.2.2 Other measures. Following the story-writing task, participants responded to a set of questions to confirm their eligibility as well as cultural experiences and exposure to robots. These served as background data for exploring relationships between narratives and the writers. An overview is provided in Table 1.

3 RESULTS

3.1 The sample groups

While all the UK and Japanese participants confirmed their identity as UK or Japanese respectively and had lived in the country of their identity as specified in 2.1., 4 in the UK and 2 in the Japanese group had at least one non-UK/-Japanese grandparent. The Japanese group (N=44) consisted mostly of young adults, either 18 to 20, or in their 20s (n=35), followed by those in their 30s (n=4), though the overall age range extended to the 60s. In the UK group (N=41), most of them were under 40 years of age: 18- 20, or in their 20s (n=17), in the 30s (n=15), the 40s (n=4), the 50s or 60s (n=5). All of the Japanese participants had seen robots before, 22 of them had interacted with one of the robots in Table 1, and 2 participants knew how to program robots. In comparison, the level of exposure was much lower in the UK group: while 32 had seen the robot in the media, 3 had never seen one, and 5 had interacted with one before. The UK group did have one person who knew how to programme robots, thus the exposure level was similar at the high end for both cultural groups. Declaration of participant gender identity were overall similar across the two cultural groups: man (24 for Japan; 19 for UK), woman (24; 23), non-binary (1; 2).

The descriptions of cultural experiences, did not suggest any particular pattern against the analysis of stories (see below). As well, there were no identifiable relationship between stories and pet ownership or pet types, or book titles or genre for either of the groups. The NARS scores were organised into the 3 subscales specified in Table 1, as well as obtaining total scores for each participant, according to the original article [9]. As has previously been reported, the scores were lower overall for the Japanese group ($mean \pm S.D. = 34.77 \pm 8.4$) than for the UK group (39.53 ± 9.55) for the total NARS score, the difference was statistically significant by rmANOVA (Supplementary materials). Despite the group level differences, distribution of the NARS scores did not suggest any relation in the analysis below for the stories.

Table 1: Background variables

Questions	Descriptions
Cross-cultural experience in family	Whether parents and grandparents were from UK/Japan
Exposure to and familiarity with robots	On general familiarity with robots, to select from: “never seen a robot” (0), seen (1), have interacted (2) or “know how to programme robots” (3); using the same scale, to indicate familiarity with Pepper/Nao/Cosmo with photographs (supplementary materials).
Age group	Across adult age-span divided in decades
Gender identity	Man/Woman/Other/decline answer
Pet ownership	Yes/No; if yes, select an animal type for a pet kept for the longest period
Cultural experience outside of Japan (Japanese group), or EU/UK (UK group)	Yes/No; if yes, which culture/via what activity/duration of experience/whether it is current or not (all in their own words) for up to three cultures.
NARS (Negative Attitude to Robots Scale)[9]	A psychometrically informed questionnaire containing 14 statements describing various anxieties around robots; responses are made to indicate the extent of agreement with each statement, from strongly disagree, disagree, undecided, agree and disagree. Total scores as well as subscales are sorted into situational, social, or emotional aspects of robot anxieties.
Favorite books	An optional question to provide up to 3 titles of books they have enjoyed reading, the author, and a short description of each.

3.2 Thematic analysis

Data from the UK and Japan were first explored separately, to gain an understanding of any trends within each country group. M.R. and J.T.G. examined the UK data to conduct thematic analysis focusing on the aspects of the stories as per OSF preregistration: the nature of the human-robot relationship; proxemic behaviours of the characters, and whether the robot was portrayed more as an agent than an object. The same was repeated for the Japanese group, which was led by M.R. as the bilingual author, later joined by Y.M., before the entire data was considered on the cross-cultural level by M.R. and Y.M. to explore similarities as well as differences between the groups. In this short report, we focus on two of the research questions: a) the nature of human—robot relationship in the stories, focussing on in- vs. out-group relationship between the human and the robot; and b) agency of the robot character in the narratives, focussing on the extent to which the robot was portrayed as a being with agency. A full set of data is available upon request, including the stories written by participants in their original languages.

At the start of the process, the stories by the participants were studied repeatedly as required, to gain a rounded understanding of narratives, as well as descriptions of the robot and the human characters, including any other characters in the narratives. This early-stage analysis identified codes around the above two research questions, from which to learn the nature of the overall human—robot relationship, as well as the role of the robot in the narratives. The information extracted during this phase was organised into short descriptions for each participant: for example, whether the relationship was mostly friendly or antagonistic, or whether the robot was portrayed as a service machine or had a more autonomous capacity. This process then fed into the subsequent stage, of organising each narrative as to whether mostly in- or out-group relationship was portrayed between the robot and the human. Similarly, aspects of

the robot in the narratives were studied and sorted into whether the level of agency for the robot seemed particularly high or low within the country group data. This process suggested that for the majority of the stories, it was possible to categorise most of the narratives into groups (e.g., high or low agency) at the level of the story rather than codes within the stories. As a qualitative study, categorisation was not forced in a binary manner, but by considering whether an aspect of a story (e.g., agency) was particularly prominent. In the Japanese group, word count for the stories offered ranged between 204 and 1069 (mean: 593.66); in the UK, the range was 110 and 559 (mean: 237.24). This is an indication of good adherence to the study instruction (see 2.2).

There were 33 stories in the Japanese data where an in-group relationship was predominant; 5 portrayed the opposite, where an out-group relationship was more dominant for the human and the robot characters. The remaining 6 cases either moved between in- and out-groups or no clear pattern could be identified. In contrast, only 8 UK stories were deemed to be mostly depicting an in-group relationship, and in 6 narratives, an antagonistic and out-group relationships were dominant, with 11 unclassifiable stories. The latter sub-group contained stories where no relationship was portrayed or developed within the narrative, because, for example, the robot seemed to remain a non-communicative machine for the human, or the human character is unable to establish any meaningful interaction with the robot. To illustrate this process, stories depicting in-group relationship included the robot being a friend, colleague or work assistant. While types of in-group relationships were similar across the UK and Japanese groups, 12 stories depicting the robot as part of a family were only found in the Japanese sample. Stories that depicted clearly out-group relationships involved antagonistic human—robot relationships involving physical fights between them, or a non-responsive robot despite human efforts. Similar number of stories took the human as the main character across the

cultural groups (UK:Japan=28: 25), though this was not consistently reflected in the content of the stories (mixed voices or perspectives).

As for the level of agency portrayed, 13 stories were deemed to depict a high level of agency in the UK sample, contrasted by 28 such cases in the Japanese sample. Examples include stories where: a robot tries to engage with humans but are repeatedly rejected, leading the robot to inform its developer that it “wishes to disappear from the world” (Japan); or, a robot takes a sarcastic stance to a friendly human, so that it “feels annoyance” towards them (UK). In contrast, in stories where agency levels of the robot were deemed low, behaviours or functions of robots were merely described without any reference to mental states or psychological implications or consequences. In support of the cross-cultural pattern, while only 5 UK stories gave the robot character a name, 15 stories in the Japanese group either had a name for the robot or were first-person narratives, as told by the robot involving the pronoun, ‘I’.

Further analysis continued, cross-examining these in-group and agency elements above for each country group before exploring them at the cross-cultural level. The process confirmed and highlighted the trend seen above for the Japanese group: over 78% of the high-agency stories in the Japanese group (n=22 out of 28) were also characterised by an in-group relationship; further, over 90% of such stories (i.e., both high-agency and in-group) depicted a friendly relationship between the human (Chihiro) and the robot (n=20 out of 22). In comparison, only 3 of the 13 high-agency stories in the UK group depicted an in-group relationship, where 2 of these stories included a friendly relationship between the robot and Sam the human character.

In both UK and Japanese groups, roughly half of the stories portrayed the robot character as a service machine: for example, a butler, driver or other types of assistants for humans: 26 in the Japanese group, and 21 in the UK group. However, while 73% of such stories depicted the robot as a high agency character in the Japanese group (19/26), there were only 3 such stories (23%) in the UK group. Table 2 below gives a summary of these. Note that the figures in the table reflect the qualitative analysis approach, thus contain overlaps across the rows of characteristics (i.e., the figures do not add up to the total sample).

Taken together, the analysis of the stories suggests that those offered by the Japanese group were more likely to involve: a higher level of agency for the robot character, an in-group as well as friendly relationship between the human and the robot. An epitome of this pattern may be seen in the 6 stories in the Japanese group where the robot was portrayed as both in-group, and high in agency, and as a member of the human’s family to care for a child, or even as a substitute parent. There was no such story in the UK group. Other themes which emerged only for the Japanese group were for example: the robot’s awareness of its limit (e.g., compared to what humans are able to do) (n=14), or stories where the robot character was capable of experiencing loneliness (or an insight into it) in relation to the human character (e.g., the robot addressed the human’s loneliness or the robot missed the human) (n=10).

4 DISCUSSION

The results are overall, in line with the pattern of other published cross-cultural studies of HRI, particularly around the amenableness

of robotic agents in Japanese people. A distinct contribution of the current study may be that it took a story-stem method, a qualitative approach to find this pattern. Instead of an experimental or a survey method where participants are instructed to engage in a particular situation or a task with a specific robot, the current study invited people to come up with such contexts themselves. The set up was designed to give people the space and freedom in which to build their own story of a fellow human of their cultural group and a robot, prompted only by a simple and relatively open-stance beginning of a story. Stories produced by people were a result of an implicit and internal process; however, in order to complete their stories, their imaginations had to be sufficiently active and detailed. In this sense, a story-stem method may present an interesting hybrid case, which may be neither explicit or implicit as a research method.

The relative freedom of story-stem method could have been a risk; for example, resulting in stories that lacked substance or details. Instead, and in support of the story-stem technique, the stories collected in the current studies met the quantitative requirement and were qualitatively rich to allow the current analysis. As a result, analysis of the stories seemed to grant a deeper insight into the often-reported friendliness of Japanese people towards robots: Japanese participants imagined the robot character to be a being rather than a machine; as someone who can become part of their everyday life; even a family member, or as a close company at a vulnerable or critical time in one’s life such as loneliness. This may relate to previous findings where Japanese people have been found to be more readily accepting of robots being a ubiquitous or intimate part of their society, for example assisting with childcare [5]. Such willingness to allow intimacy with robots which came through from Japanese people’s stories may also help account for those Japan-only themes that emerged, such as the robot lamenting their own limit—a being that is so closely a part of people’s daily life, might also grieve for not quite being a human despite being so nearly human. The current study permitted participants to choose their main character, addressing a potential limitation identified by a previous study [12]; however, this has resulted in a similar outcome, i.e., inconsistent reflection of the choice in the stories. It appears therefore, that a choice of the protagonist may not necessarily lead to a bias for participants.

In a recent study comparing samples of Dutch and Japanese people [3], the Japanese sample scored lower than the Dutch group on the NARS as in the current study; however, no cultural group difference was found in their binary sorting task to test spontaneous associations of humans and robots as an implicit measure. The current finding that Japanese people tended to treat robots more as a being than an object in their stories, may offer some bridging insight here: for a (Japanese) mind where both people and robots are considered agentic, rapid associative assessment is rather difficult without knowing what kind of context or what sort of robot is in question.

It may be worthwhile highlighting that the current study did not just invite a group of UK and Japanese people to participate, but instead, tried using more objective information such as participants’ country and duration of residency, nationalities in families, as well as participants’ own sense of cultural identity. While this seemed to us both a viable and empirically valid means to define culture

Table 2: UK-Japan comparison of characteristics of stories in terms of the robot agency, relationship (in- vs. out-group; friendly vs. antagonistic) and the role assigned for the robot

	UK (N=41)	Japan (N=44)
High agency	13	28
In-group relationship	8	33
Friendly relationship	15	35
Service robot	21	26
High agency; in-group relationship	3	22
High agency; in group & friendly relationship	2	20
High agency; service robot	3	19
High agency; in group; part of family	0	6

for the study, further discussions and refinement of recruitment methods are warranted in the HRI community.

As for limitation of the study, we do not know the relationship between the nature of the stories presented by the participants and their personal attitude or beliefs around HRI. In this sense, the study belongs to a social constructionist approach as with other previous studies employing story-stem method [12]. This reflects the remit and limit of qualitative research: the narratives generated by people are not so much about reality, but more of what an encounter with a given robot might look like on a given day. The study was entirely online during the pandemic (2021–22). While stories containing experiences of loneliness may have reflected this context, it remains only a possibility.

The current results may permit some future directions for investigating culturally sensitive design for future robots. For example, although Japanese people produced stories involving more agentic robot characters than the UK people, it may be valuable to study which purposes and context of HRI that an agentic robot is particularly welcome (or not). Modification of the story-stem may be one such way. Studies involving more cultural groups are of course warranted, but also valuable would be to further test more granular sub-groups within the UK and Japanese people respectively. Such efforts would help articulate what robotic designs are optimal for when or for whom.

Acknowledgments: The study was funded by the European Research Council, awarded to E.S.C. (ERC-StG-2015-62272- 1312 SOCIALROBOTS). We thank: Nora Holtz for the contribution to the design of this study; Jikken-Baito, managed by Atsushi Ueshima, as well as Hiroyuki Okada and Sotaro Shimada for their support with recruitment of Japanese participants. JASP 0.19.3 was used for statistical analyses. We thank all the people who participated and wrote stories for the study.

References

- [1] Victoria Clarke, Virginia Braun, Hannah Frith, and Naomi Moller. 2019. Editorial introduction to the special issue: Using story completion methods in qualitative research. , 20 pages.
- [2] Daniel Clement Dennett (Ed.). 1978. *Brainstorms: Philosophical Essays on Mind and Psychology*. Bradford Books, Cambridge, Massachusetts.
- [3] Fabiola Diana, Misako Kawahara, Isabella Saccardi, Ruud Hortensius, Akihiro Tanaka, and Mariska E Kret. 2023. A cross-cultural comparison on implicit and explicit attitudes towards artificial agents. *International Journal of Social Robotics* 15, 8 (2023), 1439–1455.
- [4] Edward T Hall. 1989. *Beyond culture*. Anchor.
- [5] Kerstin Sophie Haring, Céline Mougenot, Fuminori Ono, and Katsumi Watanabe. 2014. Cultural differences in perception and attitude towards robots. *International Journal of Affective Engineering* 13, 3 (2014), 149–157.
- [6] Ryan Blake Jackson and Tom Williams. 2021. A theory of social agency for human-robot interaction. *Frontiers in Robotics and AI* 8 (2021), 687726.
- [7] Velvetina Lim, Maki Rooksby, and Emily S Cross. 2021. Social robots on a global stage: establishing a role for culture during human-robot interaction. *International Journal of Social Robotics* 13, 6 (2021), 1307–1333.
- [8] Karl F MacDorman, Sandosh K Vasudevan, and Chin-Chang Ho. 2009. Does Japan really have robot mania? Comparing attitudes by implicit and explicit measures. *AI & society* 23 (2009), 485–510.
- [9] Tatsuya Nomura, Takayuki Kanda, and Tomohiro Suzuki. 2006. Experimental investigation into influence of negative attitudes toward robots on human-robot interaction. *Ai & Society* 20 (2006), 138–150.
- [10] Maki Rooksby, JT Goetz, Nora Holtz, and Emily S Cross. 2021. Proxi-Story study: how people in the UK and Japan imagine an encounter with a robot in response to a story stem. (2021).
- [11] Allison Tong, Peter Sainsbury, and Jonathan Craig. 2007. Consolidated criteria for reporting qualitative research (COREQ): a 32-item checklist for interviews and focus groups. *International journal for quality in health care* 19, 6 (2007), 349–357.
- [12] Giovanni Maria Traoiano, Matthew Wood, and Casper Hartevel. 2020. "And This, Kids, Is How I Met Your Mother": Consumerist, Mundane, and Uncanny Futures with Sex Robots. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*, 1–17.
- [13] Matthew Wood, Gavin Wood, and Madeline Balaam. 2017. "They're Just Tixel Pits, Man" Disputing the 'Reality' of Virtual Reality Pornography through the Story Completion Method. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 5439–5451.